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1 The infinite spine case

Throughout this section, G denotes an ordered abelian group. For $n \geq 2$ and $a \notin nG$, recall that $\mathfrak{s}_n(a)$ is the largest convex subgroup $H \subseteq G$ such that $a \notin H + nG$. If $a \in nG$, we set $\mathfrak{s}_n(a) = \emptyset$, and we regard $\emptyset$ as being smaller than every convex subgroup. We set $\mathcal{S}_n = \{\mathfrak{s}_n(a) \mid a \in G\}$. The subgroups $\mathfrak{s}_n(a)$ are uniformly definable in a [1, Lemma 2.1]. Consequently, comparisons and equalities between values of the maps $\mathfrak{s}_n$ are formulas in L_{oag} .

Lemma 1.1 ([3, Lemma 2.8(2) and Remark 2.6]). *Let p be a prime, $r \geq 1$, and $a, b \in G$.*

1. *If $0 \leq j < r$, then*

$$\mathfrak{s}_{p^r}(p^j a) = \mathfrak{s}_{p^{r-j}}(a).$$

If $j \geq r$, then $\mathfrak{s}_{p^r}(p^j a) = \emptyset$.

2. *If $\mathfrak{s}_{p^r}(a) \neq \mathfrak{s}_{p^r}(b)$, then*

$$\mathfrak{s}_{p^r}(a + b) = \max\{\mathfrak{s}_{p^r}(a), \mathfrak{s}_{p^r}(b)\}.$$

Lemma 1.2. *Let p be a prime and let $H < H'$ be nonempty elements of $\mathcal{S}_p$. There exists $a \in H'$ such that*

$$\mathfrak{s}_p(a) = H \quad \text{and} \quad \mathfrak{s}_{p^r}(a) < H'$$

for every $r \geq 1$.

Proof. Choose $b \in G$ such that $\mathfrak{s}_p(b) = H$. Since $H' > H$, the maximality of H gives $b \in H' + pG$. We may therefore write $b = a + pc$ with $a \in H'$ and $c \in G$. As $a - b \in pG$, we have $\mathfrak{s}_p(a) = \mathfrak{s}_p(b) = H$.

Fix $r \geq 1$. If $\mathfrak{s}_{p^r}(a) = \emptyset$, there is nothing to prove. Otherwise, put $K = \mathfrak{s}_{p^r}(a)$. If $K \geq H'$, then $a \in H' \subseteq K$, contrary to $a \notin K + p^r G$. Hence, $K < H'$. ■

Lemma 1.3 (Finite coding). *Let p be a prime and*

$$K_0 < K_1 < \cdots < K_N$$

be nonempty elements of $\mathcal{S}_p$. There exists $x \in G$ such that

$$\mathfrak{s}_{p^{i+1}}(x) = K_i$$

for every $0 \leq i \leq N$.

Proof. For each $j < N$, use Lemma 1.2 to choose $a_j \in G$ such that

$$\mathfrak{s}_p(a_j) = K_j \quad \text{and} \quad \mathfrak{s}_{p^r}(a_j) < K_{j+1}$$

for every $r \geq 1$. Also choose $a_N \in G$ with $\mathfrak{s}_p(a_N) = K_N$, and set

$$x = a_0 + pa_1 + \cdots + p^N a_N.$$

Fix $0 \leq i \leq N$. By Lemma 1.1,

$$\mathfrak{s}_{p^{i+1}}(p^i a_i) = \mathfrak{s}_p(a_i) = K_i.$$

If $j < i$, then

$$\mathfrak{s}_{p^{i+1}}(p^j a_j) = \mathfrak{s}_{p^{i+1-j}}(a_j) < K_{j+1} \leq K_i,$$

while, if $j > i$, then $\mathfrak{s}_{p^{i+1}}(p^j a_j) = \emptyset$. Thus, among the summands of x , the unique maximal $\mathfrak{s}_{p^{i+1}}$ -value is K_i . Repeatedly applying Lemma 1.1(2) gives $\mathfrak{s}_{p^{i+1}}(x) = K_i$. ■

Theorem 1.4. *If $\mathcal{S}_p$ is infinite for some prime p , then the dp-rank of G is $\aleph_0$.*

Proof. We work in a sufficiently saturated monster model of $\text{Th}(G)$. Since $\mathcal{S}_p$ is an infinite interpretable linear order, compactness allows us to choose elements

$$(b_{i,k})_{i,k < \omega}$$

such that all the values

$$H_{i,k} := \mathfrak{s}_p(b_{i,k})$$

are nonempty and

$$H_{i,k} < H_{j,l} \quad \text{whenever} \quad i < j \text{ or } (i = j \text{ and } k < l).$$

Indeed, every finite part of these requirements merely asks for a finite chain of distinct nonempty elements of the infinite linear order $\mathcal{S}_p$.

For a function $\eta : \omega \rightarrow \omega$, consider the partial type

$$\Sigma_\eta(x) = \{ \mathfrak{s}_{p^{i+1}}(x) = H_{i,\eta(i)} \mid i < \omega \}.$$

This type is finitely satisfiable. Indeed, given $N < \omega$, the values

$$H_{0,\eta(0)} < H_{1,\eta(1)} < \cdots < H_{N,\eta(N)}$$

form a finite increasing chain, so Lemma 1.3 realizes the first $N + 1$ conditions of Σ_η . By saturation, there is $a_\eta \in G$ realizing Σ_η .

For every $i < \omega$, let

$$\varphi_i(x; y) \quad \text{be the formula} \quad \mathfrak{s}_{p^{i+1}}(x) = \mathfrak{s}_p(y).$$

As noted above, this is an L_{oag} -formula. For all $\eta : \omega \rightarrow \omega$, $i < \omega$, and $k < \omega$, we have

$$\models \varphi_i(a_\eta; b_{i,k}) \iff H_{i,\eta(i)} = H_{i,k} \iff \eta(i) = k.$$

Hence, these formulas and parameters form an ICT-pattern of depth $\aleph_0$, and therefore the dp-rank of G is at least $\aleph_0$.

Every ordered abelian group has NIP [2]. Since L_{oag} is countable, this gives dp-rank at most $\aleph_0$. Consequently, the dp-rank of G is exactly $\aleph_0$. ■

References

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- [2] Yuri Gurevich and Peter H. Schmitt. “The theory of ordered abelian groups does not have the independence property”. *Transactions of the American Mathematical Society* 284.1 (1984), pp. 171–182. DOI: 10.1090/S0002-9947-1984-0742419-0 (cit. on p. 3).
- [3] Koki Okura. *Distality in ordered abelian groups*. 2026. arXiv: 2602.23256 [math.LO] (cit. on p. 1).